

# AI-BASED EARLY WARNING AND LANDSLIDE RISK MONITORING SYSTEM

## Technical Research Report and System Design Specification

**Problem Statement: SIH26001**

Theme: Disaster Management

Proposed technologies: IoT • LoRa • Satellite • GIS • AI/ML • Web Backend • REST APIs

This report consolidates the supplied system architecture, hardware specification, circuit design, AI/ML workflow, backend/API architecture, and dashboard interface material into a single research-paper-style technical document. The figures are placed at the relevant sections so that the document can be used as a detailed project report, design review document, or Smart India Hackathon technical submission draft.

### Abstract

Landslides in mountainous and high-rainfall regions can develop through the interaction of rainfall, soil saturation, slope movement, ground vibration, terrain characteristics, land cover, and longer-term environmental changes. The proposed system is a multi-source early-warning platform that combines IoT/LoRa sensor observations, satellite-derived information, weather data, GIS terrain variables, historical landslide information, and an AI/ML prediction engine. The supplied architecture uses an ESP32-S3-based field node, LoRa communication, a gateway/backend pipeline, REST APIs, XGBoost-based prediction, GIS risk visualization, and an early-warning layer for authorities, field teams, and a web dashboard. The system converts heterogeneous observations into a 0–100 risk score and four risk classes—Low, Moderate, High, and Critical—while an explainability layer identifies contributing factors. The report describes the proposed sensing hardware, power architecture, communication chain, data processing pipeline, machine-learning workflow, API contracts, dashboard functions, alerting mechanism, deployment considerations, and future improvements.

### Keywords

Landslide prediction, early warning system, IoT, LoRa, ESP32-S3, GIS, satellite data, SAR, NDVI, XGBoost, risk score, REST API, disaster management, sensor fusion.

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## **1. Introduction**

The proposed platform addresses the need for continuous, multi-source landslide-risk monitoring in vulnerable mountainous areas. Instead of relying on a single indicator, the system combines local field measurements with remote sensing, weather information, GIS terrain context, and historical-event information. This multi-source design is intended to support earlier detection of changing conditions and provide a common decision-support view for authorities and field teams.

The supplied design emphasizes a practical edge-to-cloud pipeline. Field nodes acquire environmental and ground-condition signals; LoRa provides long-range telemetry to a gateway; the backend receives, validates, stores, and processes observations; the AI/ML engine estimates landslide probability; GIS converts the result into a spatial risk view; and the warning engine distributes risk information to authorized users.

## **2. Problem Statement and Motivation**

The system is designed for environments where landslide risk may change rapidly due to heavy rainfall and soil saturation, while also being influenced by terrain, vegetation/land cover, surface displacement, vibration, tilt, and other environmental conditions. A useful monitoring system therefore needs to combine real-time and spatially distributed data rather than depending only on periodic manual inspection.

- Continuous sensing of rainfall, soil moisture, ground vibration, tilt, air quality, and temperature using field nodes.
- Long-range communication suitable for remote or hilly areas using LoRa.
- Integration of satellite-derived information such as SAR change and NDVI and GIS variables such as slope, elevation, and land cover.
- AI/ML-based estimation of landslide probability and conversion into an interpretable 0–100 risk score.
- Map-based visualization and alerts for authorities, field teams, and authorized dashboard users.
- Support for degraded connectivity through local gateway operation and cached data concepts shown in the dashboard design.

### 3. Objectives

1. Acquire multi-parameter environmental and ground-condition data from distributed sensor nodes.
2. Transport field observations over a long-range, low-power wireless link.
3. Fuse sensor, weather, satellite, GIS, and historical information into a unified analytical dataset.
4. Generate a landslide probability estimate using an XGBoost-based prediction engine.
5. Convert model output into a 0–100 risk score and four operational risk classes.
6. Explain the principal contributing factors behind an elevated risk estimate.
7. Visualize risk spatially and deliver actionable warnings to designated recipients.

### 4. Overall System Architecture

The high-level architecture follows a layered pipeline: field sensing and remote data acquisition → data fusion → AI/ML engine → risk score → GIS visualization → early warning → operational users.

The following supplied figure is the principal architecture representation.

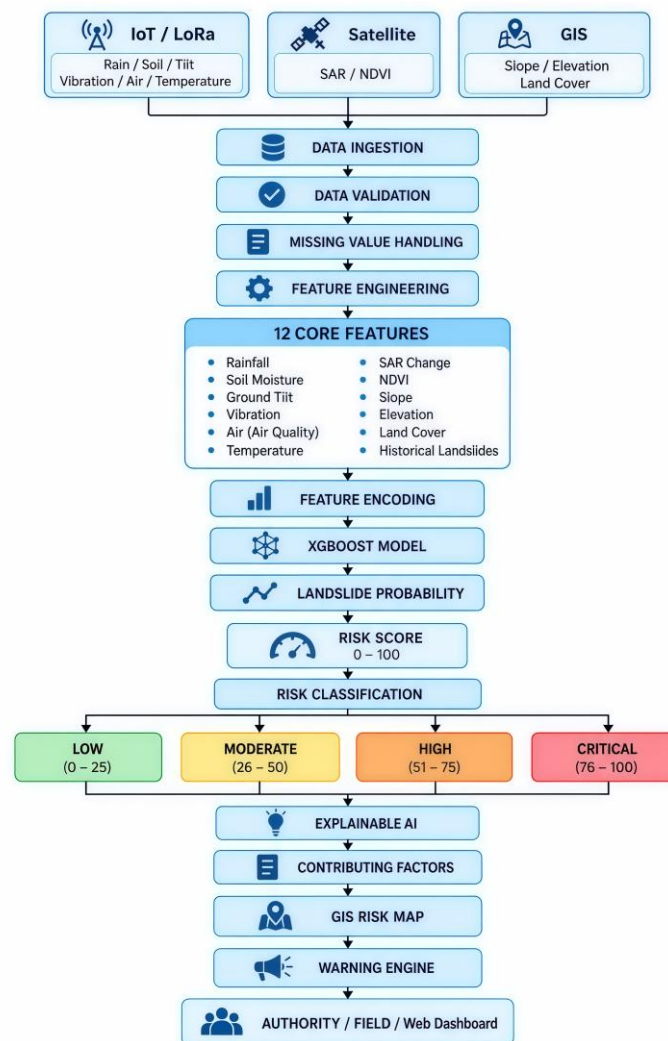
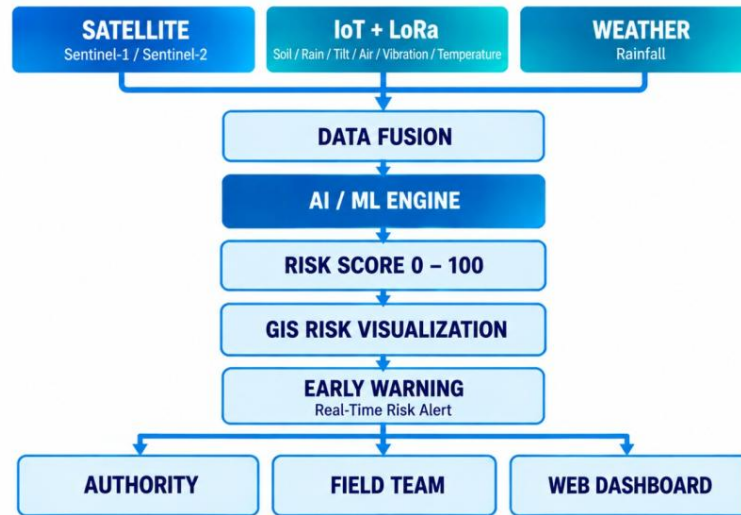


Figure 1. Complete project flow.

The flowchart is the primary analytical pipeline for the proposed system. It combines field IoT/LoRa measurements (rainfall, soil moisture, ground tilt, vibration, air quality and temperature) with satellite-derived SAR/NDVI, GIS terrain variables and historical landslide information. After validation, missing-value handling and feature engineering, the engineered feature vector is passed to the XGBoost model. The prediction is converted to a 0–100 risk score and classified as Low (0–25), Moderate (26–50), High (51–75) or Critical (76–100). Explainable-AI output identifies contributing factors before the GIS risk map and warning engine distribute the result to authorities, field teams and the web dashboard.



**Figure 2. Supplied end-to-end architecture: satellite, IoT/LoRa and weather data are fused before AI/ML prediction, GIS visualization and early warning.**

The architecture can be divided into six logical layers:

- Acquisition layer: ESP32-S3 sensor nodes, weather feeds and satellite observations.
- Communication layer: LoRa field telemetry and gateway-to-backend connectivity.
- Data layer: ingestion, validation, missing-value handling, storage and zone association.
- Analytics layer: feature engineering, feature encoding and XGBoost inference.
- Decision layer: landslide probability, 0–100 risk score, risk classification and explainability.
- Presentation/response layer: GIS risk map, early-warning engine, authority view, field-team view and web dashboard.

The complete flowchart clarifies that the prediction engine is not an isolated machine-learning component. It is positioned after data-quality controls and feature engineering so that unreliable or incomplete observations do not directly enter the prediction stage. The output is also not limited to a probability value: it is transformed into an operational risk score, accompanied by contributing-factor explanations, mapped spatially and connected to a warning workflow. This separation supports traceability from a field measurement to the final dashboard decision.

## 5. Flow Explanation — Brief

| Stage                  | Purpose                                                                                                                                                                 |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data Sources           | IoT/LoRa field sensors, satellite observations (SAR/NDVI), and GIS terrain/land-cover layers provide complementary measurements.                                        |
| Data Ingestion         | All incoming observations are collected into a common data pipeline with timestamps and location information.                                                           |
| Data Validation        | Range checks, sensor-status checks and consistency checks reduce the effect of faulty or noisy readings.                                                                |
| Missing Value Handling | Missing observations are flagged and handled using a defined imputation/fallback strategy instead of silently treating them as valid measurements.                      |
| Feature Engineering    | Raw observations are transformed into model-ready variables and aligned by location and time.                                                                           |
| 12 Core Features       | The model combines rainfall, soil moisture, ground tilt, vibration, air quality, temperature, SAR change, NDVI, slope, elevation, land cover and historical landslides. |
| Feature Encoding       | Categorical variables such as land-cover classes can be encoded into numerical representations before model inference.                                                  |
| XGBoost Model          | The architecture uses an XGBoost-based predictive stage to estimate landslide probability from the engineered feature vector.                                           |
| Landslide Probability  | The model output represents the estimated probability/risk likelihood for the monitored location and conditions.                                                        |

|                                   |                                                                                                                                         |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Risk Score                        | The probability is converted to a 0–100 score for easier operational interpretation.                                                    |
| Risk Classification               | The displayed architecture groups the score into Low (0–25), Moderate (26–50), High (51–75), and Critical (76–100).                     |
| Explainable AI                    | The system exposes contributing factors so an operator can understand which environmental/terrain variables are influencing the result. |
| GIS Risk Map                      | Risk values are spatially visualized so authorities can inspect affected zones, sensors and terrain context.                            |
| Warning Engine                    | The alert layer converts risk states and trigger conditions into actionable warnings for the operational interface.                     |
| Authority / Field / Web Dashboard | Final information is presented to authorities and field teams through a monitoring dashboard for situational awareness and response.    |

## 6. Core Features Used by the Risk Pipeline

| Feature               | Brief role                                                                                           |
|-----------------------|------------------------------------------------------------------------------------------------------|
| Rainfall              | Rain intensity/occurrence provides a key external trigger for increased slope instability.           |
| Soil Moisture         | Represents water accumulation/saturation in the soil.                                                |
| Ground Tilt           | Captures changes in slope/node inclination that may indicate movement.                               |
| Vibration             | Detects vibration or shock events around the monitoring node.                                        |
| Air Quality           | SPS30 particulate information adds environmental context to the monitoring node.                     |
| Temperature           | BME280 temperature supports environmental condition and trend analysis.                              |
| SAR Change            | Satellite SAR change information can indicate surface deformation/change over time.                  |
| NDVI                  | Vegetation index supports land-condition and vegetation-cover assessment.                            |
| Slope                 | Terrain slope from GIS/DEM is a major spatial susceptibility factor.                                 |
| Elevation             | Terrain elevation provides topographic context.                                                      |
| Land Cover            | Land-cover class helps characterize surface conditions.                                              |
| Historical Landslides | Past landslide locations/events provide historical risk context for model training and GIS analysis. |

## 7. All Hardware Components — Image Reference

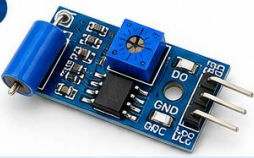
|                                                                                                                                                      |                                                                                                                                                               |                                                                                                                                                               |                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b><br><br><b>ESP32-S3 Development Board</b>                   | <b>2</b><br><br><b>Sensirion SPS30</b><br>(Particulate / Air-Quality Sensor) | <b>3</b><br><br><b>DFRobot SEN0193</b><br>(Capacitive Soil-Moisture Sensor) | <b>4</b><br><br><b>BME280</b><br>(Temperature + Humidity + Pressure) |
| <b>5</b><br><br><b>SW-420</b><br>(Vibration Sensor Module)          | <b>6</b><br><br><b>Rain Sensor Module</b><br>(Rain Detection)                | <b>7</b><br><br><b>SW-520D</b><br>(Tilt Sensor)                             | <b>8</b><br><br><b>LoRa SX1276 Module</b><br>(LoRa Communication)    |
| <b>9</b><br><br><b>Status LEDs</b><br>(Red / Green / Yellow / Blue) | <b>10</b><br><br><b>TPS62162</b><br>(3.3 V Buck Regulator)                   | <b>11</b><br><br><b>5 V Buck Regulator</b><br>(DC-DC Step Down Module)      | <b>12</b><br><br><b>SS34</b><br>(Schottky Protection Diode)          |
| <b>13</b><br><br><b>SMBJ5.0A</b><br>(TVS Protection Diode)         |                                                                                                                                                               |                                                                                                                                                               |                                                                                                                                                         |

Figure 2. Component image sheet covering all 13 hardware items in the provided component table.

## 8. Component-wise Brief Information

|                                                                                                 |                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1</b><br> | <b>1. ESP32-S3</b><br>Main edge controller. Reads sensors, performs local preprocessing, and coordinates LoRa communication.<br><br>Normal operating voltage: 3.3 V<br>Maximum / limit: 3.6 V<br>Approx. current: ~200–500 mA peak |
|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

2



## 2. SPS30

Particulate/air-quality sensor. Adds airborne particulate information to the environmental context.

Normal operating voltage: 5 V

Maximum / limit: 5.5 V

Approx. current: 55 mA avg / 80 mA max

3



## 3. SEN0193

Capacitive soil-moisture sensor. Measures soil water content, an important slope-saturation indicator.

Normal operating voltage: 3.3–5 V

Maximum / limit: 5.5 V

Approx. current: ~5 mA

4



## 4. BME280

Environmental sensor for temperature, humidity and barometric pressure.

Normal operating voltage: 3.3 V

Maximum / limit: 3.6 V

Approx. current: 0.1–1 mA

5



## 5. SW-420

Vibration sensor module. Detects vibration/shock events that can indicate ground disturbance.

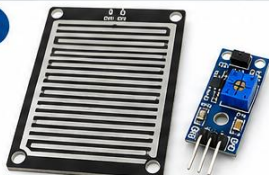
Normal operating voltage: 3.3–5 V

Maximum / limit: 5 V

Approx. current: ~15 mA

**Sensation SF350**  
(Particulate / Air-Quality Sensor)

6



## 6. Rain Sensor Module

Detects rainfall/wetness and provides a rainfall trigger signal for the risk pipeline.

Normal operating voltage: 3.3–5 V

Maximum / limit: 5 V

Approx. current: 5–20 mA

**DFROBOT SEN0195**  
(Capacitive Soil-Moisture Sensor)

7



## 7. SW-520D

Tilt sensor. Detects changes in orientation/tilt of the monitored structure or slope node.

Normal operating voltage: 3.3–5 V

Maximum / limit: 5 V

Approx. current: ~5 mA

**BME280**  
(Temperature + Humidity + Pressure)

8



## 8. LoRa SX1276

Long-range low-power radio link from field sensor nodes to a gateway/base station.

Normal operating voltage: 3.3 V

Maximum / limit: 3.7 V typ.

Approx. current: 60–120 mA TX peak

**SW-420**  
(Vibration Sensor Module)

9



## 9. Status LEDs

Visual indication for power, sensing, communication or alert states.

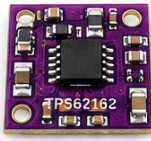
Normal operating voltage: 3.3 / 5 V

Maximum / limit: 5 V

Approx. current: 2–10 mA each

Rain Sensor Module  
(Rain Detection)

10



TPS62162

### 10. TPS62162

3.3 V switching regulator used to provide a regulated rail for the controller/sensors.

Normal operating voltage: 3–17 V input

Maximum / limit: 17 V max input

Approx. current: Up to 1 A output

SW-520D  
(Tilt Sensor)

11



5 V Buck Regulator

### 11. 5 V Buck Regulator

DC-DC converter that derives a regulated 5 V rail from the battery/system supply.

Normal operating voltage: From battery

Maximum / limit: Module-dependent

Approx. current: Up to 2 A recommended

LoRa SX1276 Module  
(LoRa Communication)

12



SS34

### 12. SS34

Schottky protection diode used for reverse-polarity/current-path protection.

Normal operating voltage: System rail

Maximum / limit: 40 V max reverse

Approx. current: ~3 A avg

(Red / Green / Yellow / Blue)

13



SMBJ5.0A  
(TVS Protection Diode)

### 13. SMBJ5.0A TVS

Transient-voltage suppressor for protecting the 5 V rail against voltage spikes.

Normal operating voltage: 5 V rail

Maximum / limit: 9.2 V clamping

Approx. current: —

## 9. Hardware-to-Flow Mapping

| Hardware / Module                 | Flow Stage               | Contribution                                                                           |
|-----------------------------------|--------------------------|----------------------------------------------------------------------------------------|
| ESP32-S3 + sensors                | IoT / LoRa input layer   | Collects local environmental and motion measurements.                                  |
| SPS30 + BME280                    | Air / Temperature inputs | Provides environmental context for the monitored site.                                 |
| SEN0193 + Rain + SW-420 + SW-520D | Ground/trigger inputs    | Captures soil moisture, rainfall, vibration and tilt signals.                          |
| SX1276 LoRa                       | Communication            | Transfers field-node data to a gateway/base station over a long-range low-power link.  |
| TPS62162 + 5 V Buck               | Power regulation         | Creates regulated supply rails for digital and sensor loads.                           |
| SS34 + SMBJ5.0A                   | Protection               | Provides reverse-polarity and transient protection at the power input/rail.            |
| GIS + Satellite                   | External spatial data    | Adds terrain, land-cover, NDVI and SAR-change information.                             |
| XGBoost + Explainable AI          | Risk engine              | Transforms the feature vector into probability, score, class and contributing factors. |
| GIS Risk Map + Warning Engine     | Decision support         | Turns model output into spatial visualization and operational warnings.                |

## 10. Technical Notes

- The voltage/current values in the component section reproduce the values supplied in the user's reference table; final PCB design should be checked against the exact vendor part number and module revision.
- The SPS30 is a particulate-matter sensor using optical/laser scattering and supports I<sup>2</sup>C/UART communication; its documented supply range is 4.5–5.5 V.
- The BME280 measures temperature, humidity and pressure and supports I<sup>2</sup>C/SPI; the sensor itself is a low-power environmental sensor.

## 11. Data Sources and Data Fusion

### 11.1 IoT / LoRa Data

The supplied architecture identifies rainfall, soil moisture, ground tilt, vibration, air quality and temperature as the core field measurements. These observations provide localized information about the conditions at or near a monitored zone.

### 11.2 Satellite Data

The architecture specifies Sentinel-1 / Sentinel-2 as satellite sources and identifies SAR and NDVI as analytical inputs. SAR change can provide information related to surface displacement or temporal surface change, while NDVI provides vegetation-related context. The exact satellite processing algorithm and revisit-dependent update schedule are not specified in the supplied material and therefore remain implementation parameters.

### 11.3 Weather and GIS Data

Weather input is represented primarily by rainfall. GIS input includes slope, elevation and land cover. Historical landslide information is also listed among the 12 core features in the supplied AI/ML workflow.

### 11.4 Twelve Core Features

| No. | Feature           | Role in analytical pipeline            |
|-----|-------------------|----------------------------------------|
| 1   | Rainfall          | Rainfall intensity/accumulation signal |
| 2   | Soil Moisture     | Local soil saturation indicator        |
| 3   | Ground Tilt       | Slope/inclination change               |
| 4   | Vibration         | Ground movement/vibration signal       |
| 5   | Air (Air Quality) | Air-quality/environmental indicator    |
| 6   | Temperature       | Ambient environmental condition        |
| 7   | SAR Change        | Satellite surface-change indicator     |
| 8   | NDVI              | Vegetation condition indicator         |
| 9   | Slope             | Terrain steepness                      |
| 10  | Elevation         | Terrain height                         |

## 11.5 Actual Working of Data Sources in the Risk Flow

How field, satellite and GIS data are collected, prepared and passed into the risk-analysis pipeline.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>The image shows a software interface for IoT / LoRa data collection. On the left, there is a photograph of a field station with a solar panel and a LoRa antenna. On the right, a list of data inputs is displayed: Rainfall (with a cloud icon), Soil Moisture (with a plant icon), and Ground Tilt (with a level icon). The title 'IoT / LoRa' is at the top.</p>          | <p><b>IoT / LoRa</b></p> <p>Inputs: Rainfall • Soil Moisture • Ground Tilt</p> <p>Actual working:</p> <p>Field sensors continuously observe local conditions. The ESP32-S3 reads the sensor signals, packages the observations with node/time information, and the LoRa link sends the field data to the receiving gateway/backend. These measurements form the real-time ground-observation input.</p> |
|  <p>The image shows a software interface for Satellite data collection. On the left, there is a photograph of a satellite in orbit. On the right, a list of data inputs is displayed: SAR Change (with a grayscale satellite image icon) and NDVI (with a green vegetation map icon). The title 'Satellite' is at the top.</p>                                                 | <p><b>Satellite</b></p> <p>Inputs: SAR Change • NDVI</p> <p>Actual working:</p> <p>Satellite products are obtained for the monitored region. SAR observations are processed to identify surface-change information, while NDVI is derived from optical imagery to represent vegetation condition.</p>                                                                                                   |
|  <p>The image shows a software interface for GIS data collection. On the left, there is a 3D visualization of terrain layers. On the right, a list of data inputs is displayed: Slope (with a triangle icon), Elevation (with a color-coded map icon), Land Cover (with a green map icon), and Historical Landslides (with a red map icon). The title 'GIS' is at the top.</p> | <p><b>GIS</b></p> <p>Inputs: Slope • Elevation • Land Cover • Historical Landslides</p> <p>Actual working:</p> <p>GIS/DEM and historical spatial layers provide terrain and background context. Slope and elevation are derived from terrain data, land-cover information describes surface classes.</p>                                                                                                |

## 12. Actual Working Flow

| Stage                     | What happens                                                                                     |
|---------------------------|--------------------------------------------------------------------------------------------------|
| 1. Field Observation      | Rainfall, soil moisture and ground tilt are measured at the field node.                          |
| 2. Local Processing       | ESP32-S3 reads and organizes the sensor observations.                                            |
| 3. LoRa Transmission      | Field observations are transmitted over the LoRa communication link to the receiving gateway.    |
| 4. Satellite Processing   | SAR change and NDVI information are prepared for the same monitored area.                        |
| 5. GIS Processing         | Slope, elevation, land cover and historical landslide layers are prepared and spatially aligned. |
| 6. Data Ingestion         | IoT/LoRa, satellite and GIS outputs enter the common data-ingestion stage.                       |
| 7. Data Validation        | Incoming values are checked for quality, consistency and usable ranges.                          |
| 8. Missing-Value Handling | Missing or unavailable observations are identified and handled before model processing.          |
| 9. Feature Engineering    | The source observations are transformed into the feature set used by the risk model.             |

## 13. Role of Each Data Source in the Final Risk Model

| Source     | Role                              | Why it is used                                                  |
|------------|-----------------------------------|-----------------------------------------------------------------|
| IoT / LoRa | Real-time local conditions        | Captures what is happening at the monitored field location now. |
| Satellite  | Regional / surface-change context | Adds remotely observed SAR-change and vegetation information.   |
| GIS        | Terrain and historical context    | Adds stable spatial factors and past landslide information.     |

## 14. Field Hardware and Sensor Node

The supplied PCB design identifies an ESP32-S3 as the main controller and an RFM95W LoRa module as the long-range communication module. The PCB is annotated with a 150 mm × 70 mm board dimension. The design includes sensor connectors, power conversion, protection components and status LEDs.

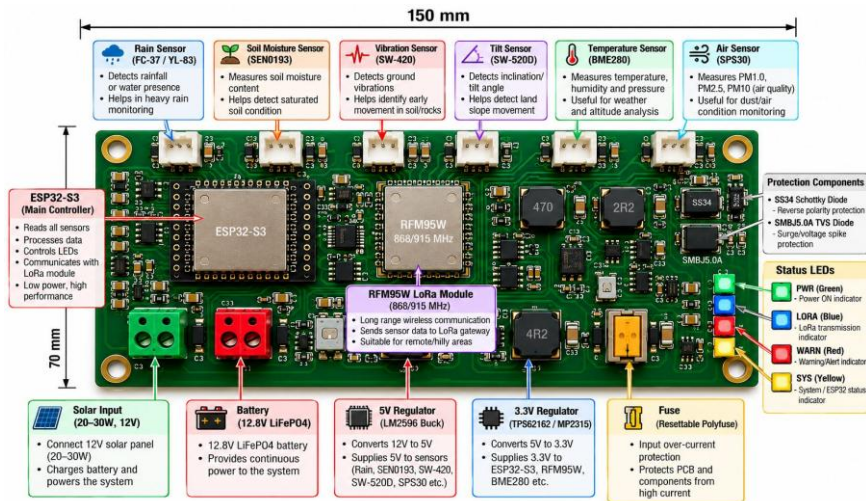


Figure 3. Project PCB architecture.

The PCB-level design shows the field node as a compact integrated acquisition and communication unit. The ESP32-S3 acts as the main controller and interfaces with rainfall, soil-moisture, vibration, tilt, BME280 environmental and SPS30 air-quality sensing. The RFM95W provides long-range telemetry. The power section accepts a 12 V solar/battery source and generates regulated 5 V and 3.3 V rails. Reverse-polarity, surge and over-current protection are included, together with PWR, LoRa, warning and system-status LEDs. The board annotation indicates an approximate 150 mm × 70 mm PCB envelope.

### 3. Detailed Hardware Specifications

| Component           | Standards Model           | Primary Technical Role               | Key Specification                                          |
|---------------------|---------------------------|--------------------------------------|------------------------------------------------------------|
| Microcontroller     | ESP32-WROOM-32D           | Edge data processing & power control | Dual-core 240MHz, Wi-Fi & BLE, Deep Sleep support          |
| Tilt & Acceleration | MPU6050                   | Monitors slope angular shifts        | 3-axis gyroscope, 3-axis accelerometer, I2C protocol       |
| Soil Moisture       | Capacitive V1.2           | Tracks volumetric water content      | Corrosion-resistant design, Analog voltage output          |
| Rainfall Sensor     | Optical / Tipping Module  | Measures real-time precipitation     | High sensitivity rain drops sensor, Digital/Analog outputs |
| Vibration Sensor    | SW-420                    | Detects sudden underground shifts    | High sensitivity, Adjustable threshold comparator          |
| Temperature         | DS18B20 (Waterproof)      | Monitors ambient/soil temperature    | Stainless steel probe, 1-Wire interface, -55°C to +125°C   |
| Long Range Wireless | SX1278 LoRa Transceiver   | Node-to-Gateway telemetry            | 860 / 915 MHz frequency band, up to 15km range             |
| Cloud Connectivity  | SIM7000G-H (4G)           | Gateway-to-cloud data & SMS          | Global LTE-TDD/LTE-FDD, GPRS support, SMS API              |
| Power Storage       | 18650 Li-ion Battery pack | Provides continuous outdoor power    | 3.7V nominal voltage, 2600mAh, combined capacity           |
| Solar Harvester     | Monocrystalline Panel     | Charges battery automatically        | 5V output, 3W maximum power rating, weatherproof           |
| System Enclosure    | IP67 ABS Plastic Box      | Protects circuit from weather        | Dustproof, heavy-duty, waterproof, integrated cable glands |

Figure 4. Supplied detailed hardware specification table (reference configuration).

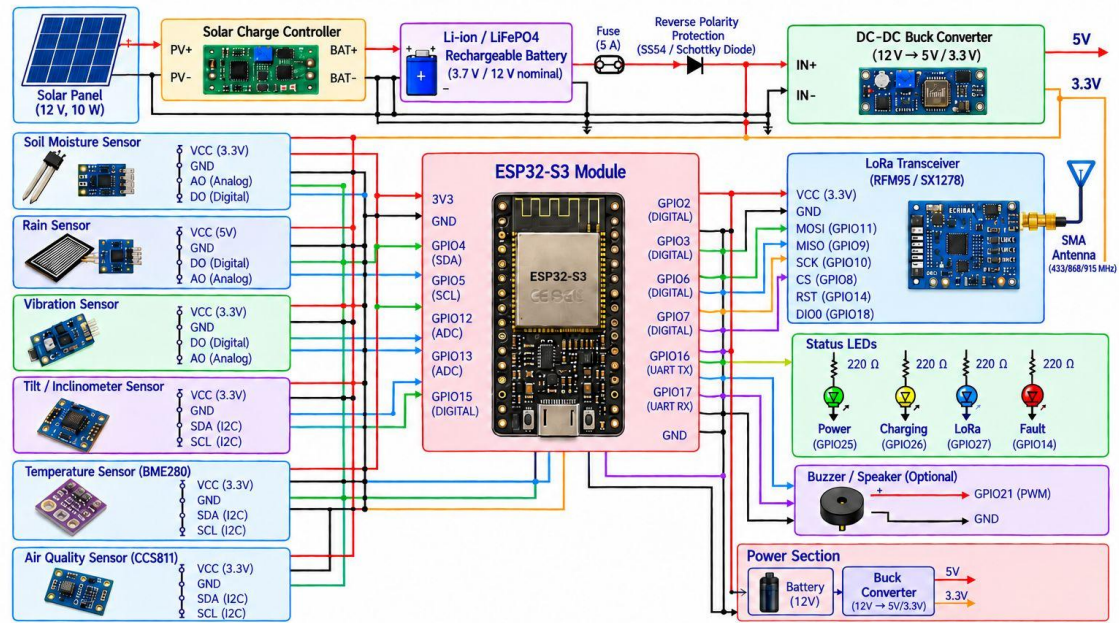


Figure 5. Supplied circuit-level architecture showing solar charging, battery, DC-DC conversion, ESP32-S3, sensors, LoRa, LEDs and optional buzzer.

## 14.1 Current PCB-Level Component Set

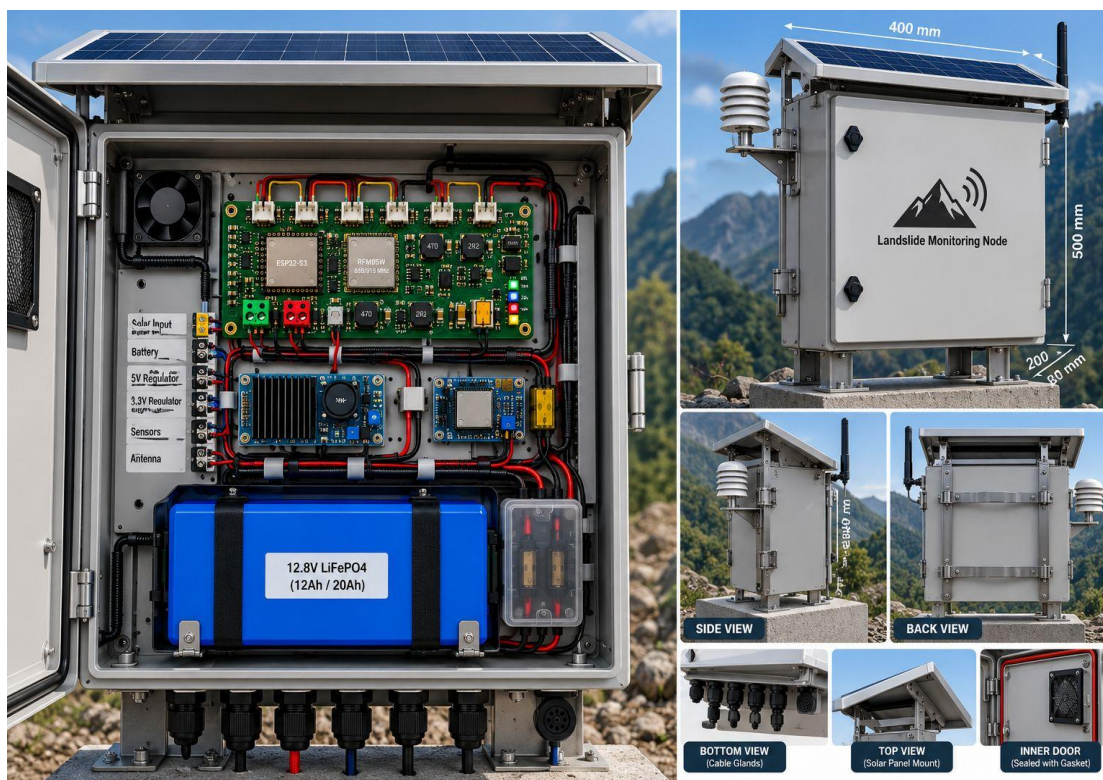
The PCB illustration supplied with the project identifies the following sensor/controller set: ESP32-S3 main controller; RFM95W 868/915 MHz LoRa module; FC-37/YL-83 rain sensor; SEN0193 soil-moisture sensor; SW-420 vibration sensor; SW-520D tilt sensor; BME280 temperature/humidity/pressure sensor; and SPS30 air-quality/particulate sensor. It also identifies SS34 reverse-polarity protection, SMBJ5.0A TVS protection, status LEDs, a resettable fuse, 5 V buck conversion and 3.3 V regulation.

- ESP32-S3: reads sensors, performs local processing, controls indicators and communicates with the LoRa module.
- RFM95W: long-range wireless telemetry at the indicated 868/915 MHz configuration.
- Rain sensor: rainfall/water-presence monitoring.
- SEN0193: soil-moisture measurement and saturated-soil indication.
- SW-420: vibration detection for ground movement indication.
- SW-520D: inclination/tilt monitoring.
- BME280: temperature, humidity and pressure measurement.
- SPS30: particulate/air-quality monitoring.

## 14.2 Supplied Reference Hardware Variant

A second supplied hardware table shows an alternative/reference configuration containing ESP32-WROOM-32D, MPU6050, Capacitive V1.2 soil-moisture sensing, an optical/tipping rainfall module, SW-420, DS18B20, SX1276 LoRa, SIM7600H-4G, an 18650 Li-ion battery pack, a monocrystalline solar panel and an IP67 ABS enclosure. Because the supplied materials contain both configurations, this report treats that table as a reference variant rather than silently replacing the PCB-level configuration.

Before fabrication, the team should freeze one bill of materials and one electrical interface specification. In particular, the selected temperature, air-quality, tilt, controller and LoRa modules should be consistent across firmware, schematic, PCB, enclosure and software documentation.



**Figure 6.** Field enclosure concept: solar-powered, sealed landslide monitoring node with internal PCB, battery, regulators, protection and external sensor/antenna interfaces.

## 14.3 Field Deployment and Mechanical Design

The field node is intended to operate continuously in exposed, high-humidity and rainfall-prone environments. The mechanical arrangement should therefore maintain sensor exposure where measurement requires it while keeping the electronics compartment sealed against water, dust and accidental cable strain. The solar panel is mounted above the electronics enclosure to provide an unobstructed charging surface. The LoRa antenna is mounted externally for improved radio performance, while cable glands provide defined entry points for sensor, power and auxiliary connections.

For installation, the enclosure should be mechanically anchored to a stable foundation or suitable terrain support. Sensor placement should be selected so that rainfall, soil moisture, vibration and tilt measurements represent the monitored slope rather than the enclosure itself. The final mechanical design should also account for lightning exposure, drainage, service access, battery replacement and safe maintenance.

The enclosure concept translates the electronic design into a deployable field node. The illustrated housing provides a protected internal mounting area for the controller PCB, power converters, battery and fuse/protection components, while cable glands provide controlled external connections. A top-mounted solar panel supports autonomous energy generation, and the external antenna and environmental sensor housing allow wireless communication and representative atmospheric measurements. The supplied concept labels an approximately 400 mm wide × 500 mm high enclosure body with a roughly 200 mm depth and an additional 80 mm mounting/base indication.

## **15. Power Supply and Protection Architecture**

The supplied circuit uses solar generation, a charge controller, a rechargeable battery, fuse/reverse-polarity protection and DC-DC conversion to create the rails required by the controller and sensors. The PCB annotation specifies a 20–30 W, 12 V solar input and a 12.8 V LiFePO<sub>4</sub> battery in the main PCB concept.

- Solar input: the PCB annotation specifies a 12 V, 20–30 W solar panel input.
- Battery: the PCB annotation specifies a 12.8 V LiFePO<sub>4</sub> battery.
- 5 V rail: generated using a buck regulator for sensors/modules requiring 5 V.
- 3.3 V rail: generated for ESP32-S3, BME280, RFM95W and other 3.3 V electronics.
- Fuse/polyfuse: provides input over-current protection.
- SS34 Schottky diode: reverse-polarity protection.
- SMBJ5.0A TVS: surge/transient protection on the indicated rail.
- Status LEDs: power, LoRa activity, warning and system status indication.

## **16. Communication and LoRa Network**

The communication architecture is designed around a field node → LoRa gateway → backend server path. The field node performs local acquisition and sends compact telemetry. A gateway receives LoRa packets and forwards them through an available backhaul such as the Internet/4G connection represented in the dashboard concept.

- Sensor node: reads and packages local measurements.
- LoRa link: transports telemetry over the long-range wireless network.

- Gateway: aggregates packets and forwards them to the backend.
- Backend: validates, stores, processes and exposes the data through APIs.
- AI/ML engine: receives engineered features and returns a risk estimate.
- Dashboard: presents current status, map risk, alerts and connectivity.

The dashboard mock-up also represents connectivity status such as connected nodes, offline nodes, packet delivery, RSSI and SNR. These are useful operational metrics because an apparent absence of risk data may otherwise be confused with a genuinely low-risk condition.

## 17. AI/ML Prediction Methodology

The supplied AI/ML flow defines the following sequence: data ingestion → data validation → missing-value handling → feature engineering → twelve core features → feature encoding → XGBoost model → landslide probability → risk score → risk classification → explainable AI → contributing factors → GIS risk map → warning engine.

XGBoost is positioned as the principal supervised learning model in the supplied design. The model consumes engineered numerical/categorical representations of the multi-source feature set and produces a landslide-risk probability or classification signal. The exact training dataset, labels, class balance, hyperparameters, validation split and calibrated probability method are not specified in the supplied materials; these should be documented during implementation and experimentation.

The supplied flowchart identifies twelve core features: rainfall, soil moisture, ground tilt, vibration, air quality, temperature, SAR change, NDVI, slope, elevation, land cover and historical landslides. These features represent complementary signals: rainfall and soil moisture capture hydrometeorological loading; tilt and vibration capture local ground response; temperature and air-quality observations provide environmental context; and SAR, NDVI, slope, elevation, land cover and historical landslide information provide remote-sensing, terrain and historical context. The exact feature scaling, training dataset, labels and hyperparameters are not specified in the supplied material and should therefore be finalized during model development.

### 17.1 Data Preprocessing

1. Data ingestion: receive timestamped observations and associate each record with a sensor node and monitored zone.
2. Validation: verify range, type, timestamp, sensor status and expected sampling behaviour.
3. Missing-value handling: detect missing or stale readings and apply the selected strategy, such as imputation, carry-forward within a defined window, or feature masking.
4. Feature engineering: derive temporal and spatial summaries such as recent rainfall accumulation, soil-moisture trend and changes from a baseline where supported by the available data.

5. Feature encoding: convert categorical fields and zone metadata into model-compatible representations.
6. Inference: submit the prepared feature vector to XGBoost.
7. Post-processing: convert model output to a 0–100 risk score and operational risk class.

## 17.2 Risk Classification

| Risk class | Score range | Operational interpretation                                                              |
|------------|-------------|-----------------------------------------------------------------------------------------|
| Low        | 0–25        | Lower estimated risk; continue routine monitoring.                                      |
| Moderate   | 26–50       | Increased attention and continued monitoring.                                           |
| High       | 51–75       | Elevated risk; authority/field review and closer monitoring.                            |
| Critical   | 76–100      | Very high estimated risk; trigger configured warning and field verification procedures. |

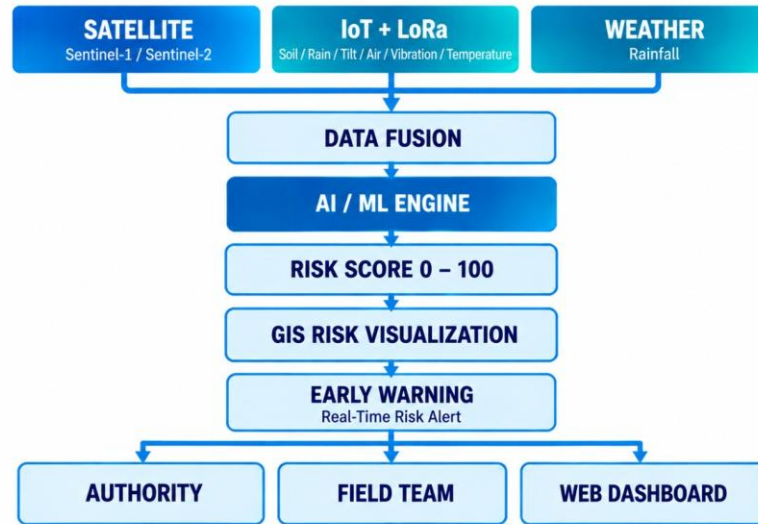
## 18. Risk Scoring and Explainable AI

The design represents a continuous 0–100 risk score followed by four risk classes. The explainability layer is intended to answer the operational question: 'Why is the risk increasing?' The supplied dashboard concept shows rainfall, soil moisture, ground movement, slope, SAR change and historical risk as contributing factors.

For implementation, the explanation should be generated from the actual model and input record rather than hard-coded dashboard percentages. Appropriate model explanation techniques can include feature contribution methods compatible with tree-based models. The explanation should clearly distinguish model contribution from physical causation: a feature that contributes strongly to a model prediction is not, by itself, proof that the feature caused a landslide.

## 19. Backend Architecture and REST APIs

The supplied backend architecture is: ESP32 sensors → LoRa gateway → backend server → REST APIs → dashboard. The backend acts as the integration layer between field telemetry, AI/ML inference, zone data, alerts and the user interface.



**Figure 7. Overall architecture figure reused as the system-level reference for backend integration and data movement.**

## 19.1 Core API Set

| API                    | Method | Endpoint                 | Purpose                                                                    |
|------------------------|--------|--------------------------|----------------------------------------------------------------------------|
| Sensor Data API        | POST   | /api/sensor-data         | Receive real-time ESP32/IoT readings.                                      |
| XGBoost Prediction API | POST   | /api/predict             | Send engineered readings/features to the XGBoost prediction service.       |
| Dashboard API          | GET    | /api/dashboard/{zone_id} | Return current readings, status and risk information for a monitored zone. |
| Alert API              | POST   | /api/alerts              | Create a landslide warning/alert record.                                   |
| Zone API               | GET    | /api/zones               | Return monitored zones and zone metadata.                                  |
| Authentication API     | POST   | /api/login               | Authenticate authorized dashboard users.                                   |

## 19.2 Suggested Request/Response Structure

The exact payload schema is not supplied, so the following is a design template rather than a claimed final contract. A sensor-data payload should contain a node identifier, zone identifier, timestamp and sensor values. A prediction request should contain the engineered feature vector and relevant zone/time metadata. A dashboard response should combine current sensor status, risk score/class, recent alerts, connectivity and map-ready information.

## 19.3 Backend Processing Sequence

1. Authenticate the caller or verify the gateway/device identity.
2. Receive sensor telemetry through the sensor-data endpoint.
3. Validate schema, timestamp and sensor ranges.
4. Persist raw observations for traceability and later model development.
5. Generate the current feature vector using the configured feature-engineering pipeline.
6. Invoke the XGBoost prediction service.
7. Store prediction, risk score, class and explanation metadata.
8. Evaluate alert thresholds and create an alert when the configured warning condition is satisfied.
9. Expose the latest zone state to the dashboard API.
10. Maintain alert history and zone-level status for operational review.

## 20. GIS Risk Visualization and Early Warning

The GIS layer converts numerical risk into a spatial decision-support view. Each monitored zone can be displayed using a risk class, while additional layers can represent terrain, sensor locations, satellite-derived change and other contextual information. The early-warning engine receives the classified risk state and distributes a configured alert to the appropriate operational audience.

- Zone-based risk visualization.
- Sensor/node status overlay.
- Risk-class filtering.
- Terrain and land-cover context.
- Historical and recent alert timelines.
- Authority and field-team notification workflow.
- Dashboard display of connectivity and data freshness.

## 21. Web Dashboard Design

The supplied dashboard concepts provide a dark, operational interface intended for continuous monitoring. The screens include a regional GIS risk map, live sensor cards, early-warning cards, connectivity status, AI prediction metrics, explainability, region summaries and a rainfall/landslide correlation area.

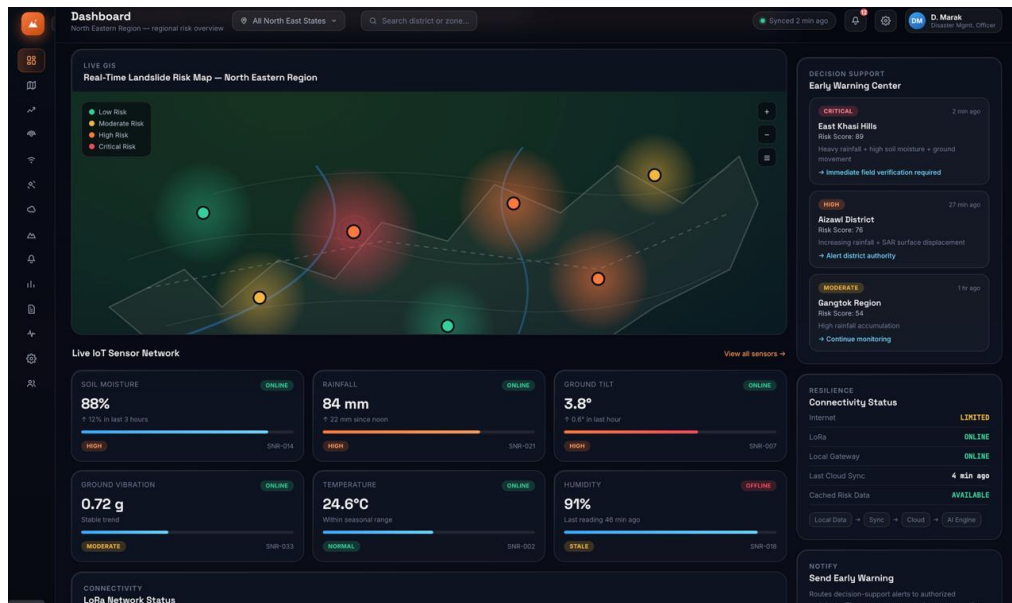


Figure 8. Supplied dashboard concept: real-time GIS risk map, live IoT readings and early-warning center.

The first dashboard view is organized around immediate situational awareness. The map shows risk locations, while sensor cards provide soil moisture, rainfall, ground tilt, vibration, temperature and humidity/other environmental information. The decision-support panel highlights active warnings and the connectivity panel indicates Internet, LoRa and gateway status.

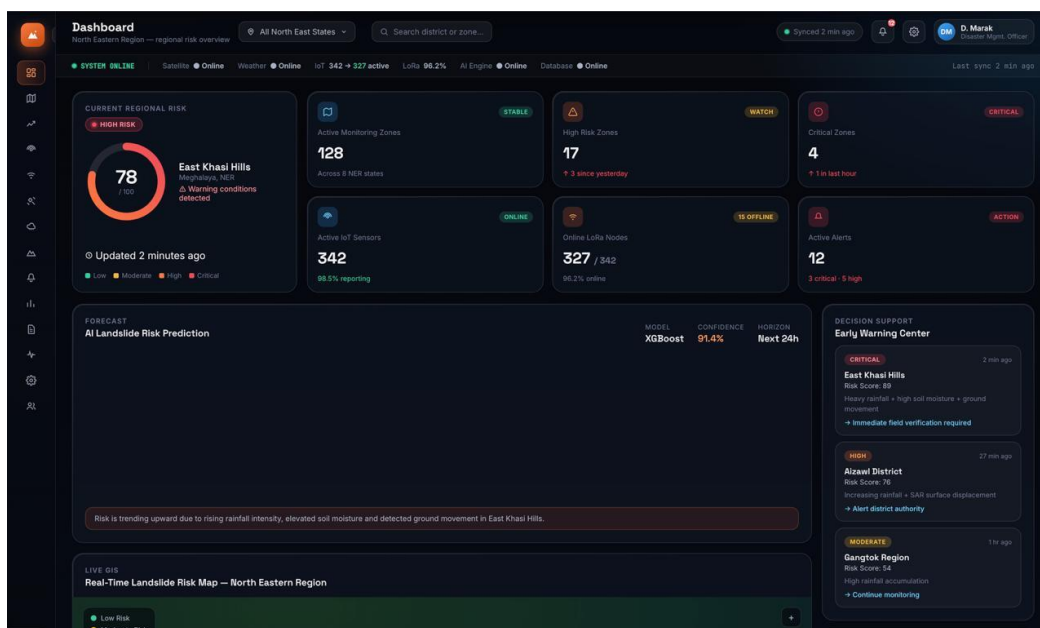


Figure 9. Supplied dashboard concept: regional risk summary, active monitoring zones, AI prediction area and early-warning panel.

The regional summary view provides high-level counts such as active monitoring zones, high-risk zones, critical zones, active IoT sensors, LoRa node status and active alerts. The AI prediction section presents the model name and confidence-style display used in the mock-up. Such values should be connected to measured evaluation results before being presented as validated performance.

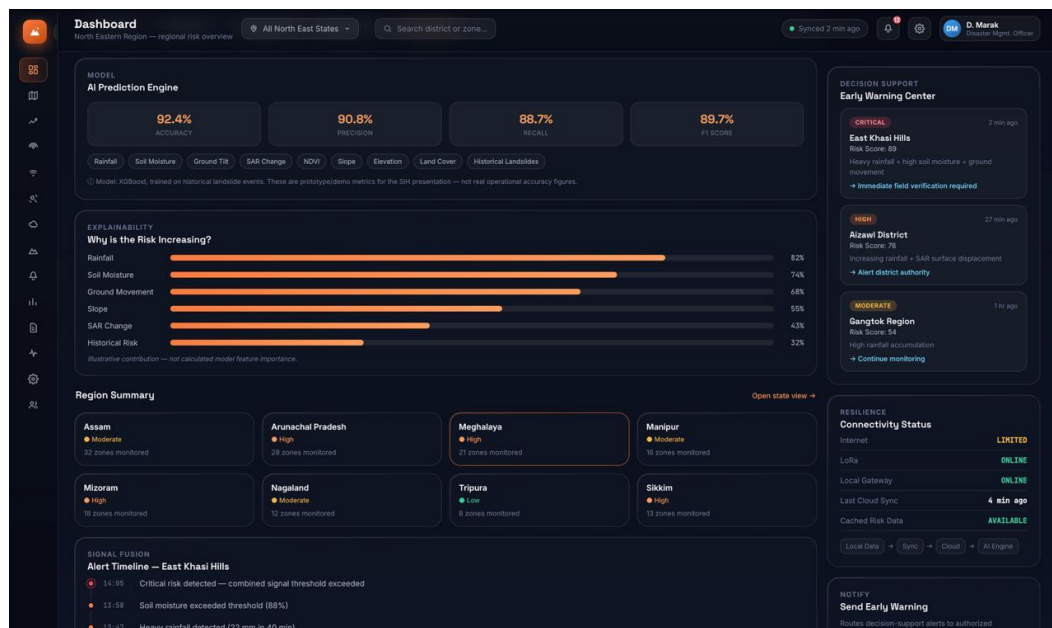


Figure 10. Supplied dashboard concept: AI prediction engine, explainability panel, region summary and alert timeline.

The AI analytics screen is designed to show accuracy, precision, recall and F1-style indicators together with the input feature set. The supplied mock-up explicitly labels its metrics as prototype/demo values rather than operational accuracy figures. In a deployed report, these metrics should be replaced by results from a documented validation experiment.

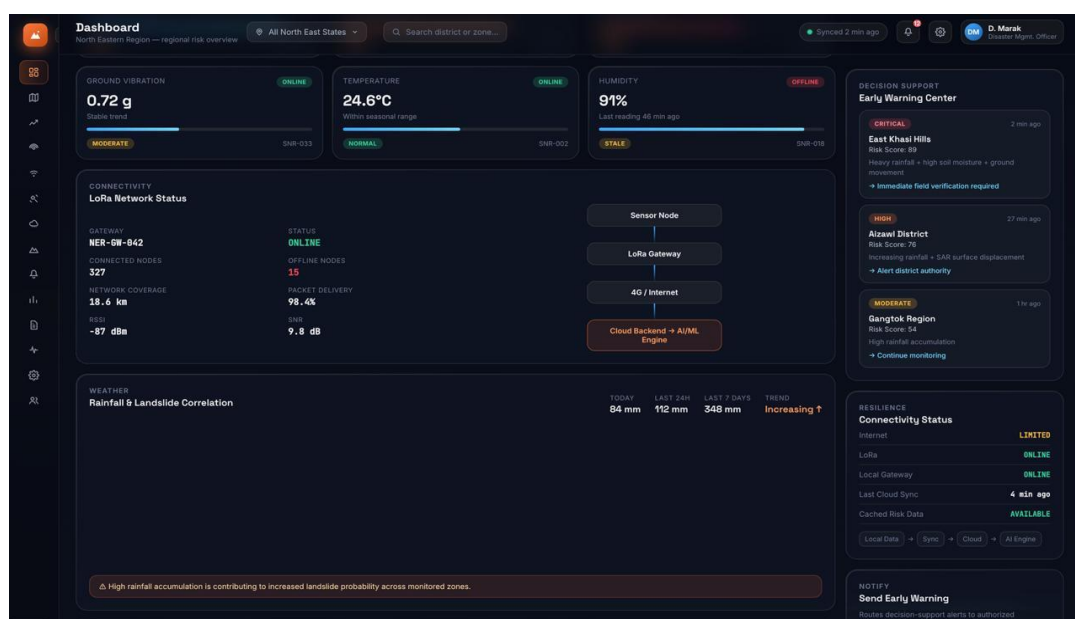


Figure 11. Supplied dashboard concept: LoRa network status, weather/rainfall context, connectivity resilience and warning panel.

The connectivity and weather view provides a network health perspective. It displays gateway state, connected/offline nodes, network coverage, RSSI, SNR, packet delivery and cloud/AI processing path. A rainfall-and-landslide correlation section can help operators understand recent environmental context.

## 22. Recommended Dashboard Modules

| Module               | Key information                                                    | Operational purpose             |
|----------------------|--------------------------------------------------------------------|---------------------------------|
| Regional Risk        | Current risk score/class and zone counts                           | Fast situational awareness      |
| Live GIS Map         | Zone risk, sensors, terrain/context                                | Spatial decision support        |
| Sensor Cards         | Rainfall, soil moisture, tilt, vibration, temperature, air quality | Current field condition         |
| AI Prediction        | Prediction output and model status                                 | Risk estimation                 |
| Explainability       | Contributing factors                                               | Interpretation of elevated risk |
| Early Warning Center | Critical/high/moderate alerts                                      | Prioritize response             |
| Connectivity         | LoRa/gateway/Internet status                                       | Data reliability monitoring     |
| Alert Timeline       | Recent threshold events                                            | Incident reconstruction         |
| Region Summary       | State/region zone overview                                         | Regional coordination           |

## 23. End-to-End Operational Workflow

1. Field sensors continuously acquire local environmental and ground-condition measurements.
2. ESP32-S3 reads and packages the observations.
3. LoRa transmits the telemetry to the gateway.
4. The gateway forwards data to the backend when backhaul connectivity is available.
5. The backend validates incoming data and handles missing/stale values.
6. Satellite, weather and GIS data are associated with the relevant monitored zone.
7. Feature engineering creates the 12-feature model input.
8. XGBoost generates a landslide-probability/risk estimate.
9. The system converts the estimate into a 0–100 risk score.
10. The score is classified as Low, Moderate, High or Critical.
11. Explainability identifies the principal model-contributing factors.
12. GIS visualizes the risk spatially.
13. The warning engine generates an alert according to configured thresholds.

14. Authorities, field teams and dashboard users receive the relevant operational information.
15. All readings, predictions and alerts can be retained for audit, evaluation and future model training.

## **24. Security, Reliability and Offline Resilience**

Because the platform is intended for disaster-management decision support, security and data reliability should be treated as system requirements rather than optional features.

- Authentication: protect dashboard access using the supplied /api/login endpoint and role-based authorization.
- Device identity: assign unique identities to sensor nodes and gateways.
- Transport security: use encrypted transport between gateway/backhaul and backend services.
- Input validation: reject malformed, out-of-range or stale sensor payloads.
- Audit trail: preserve timestamps, node IDs, prediction versions and alert events.
- Data freshness: expose last-seen timestamps so operators can distinguish current readings from stale data.
- Gateway buffering: retain unsent telemetry locally when Internet/backhaul connectivity is unavailable, then forward it after recovery.
- Alert reliability: avoid repeated duplicate alerts by using event IDs, cooldown periods and state transitions.
- Model versioning: record the model version used for each prediction so historical decisions remain traceable.

## **25. Testing and Evaluation Plan**

The supplied materials define the system architecture and prototype interface but do not provide a completed empirical validation dataset. The following test plan should therefore be executed before claiming field-level performance.

| Test area          | Example test                                                   | Expected result                                     | Evidence             |
|--------------------|----------------------------------------------------------------|-----------------------------------------------------|----------------------|
| Sensor validation  | Compare sensor readings against reference instruments          | Acceptable error within selected limits             | Calibration records  |
| LoRa communication | Measure packet delivery and range under representative terrain | Reliable telemetry within design target             | RSSI/SNR/packet logs |
| Power endurance    | Operate node across day/night cycle                            | Stable operation without brownout                   | Voltage/current logs |
| API testing        | Send valid/invalid payloads to every endpoint                  | Correct status codes and validation                 | Automated API tests  |
| ML validation      | Train/test using labeled historical events                     | Documented precision/recall/F1 and confusion matrix | Experiment report    |
| Alert testing      | Inject configured high-risk conditions                         | Correct warning state and notification              | Alert logs           |
| Dashboard testing  | Verify current readings, stale data and map updates            | Accurate UI state                                   | UI test checklist    |
| Recovery testing   | Disconnect Internet/backhaul                                   | Local buffering and later synchronization           | Recovery logs        |

## 26. Limitations and Assumptions

- The supplied material is a proposed/prototype architecture; it does not by itself establish field-validated prediction accuracy.
- The exact training dataset and labeling strategy for XGBoost are not included in the supplied material.
- Satellite preprocessing, update frequency and spatial resolution are implementation details that must be finalized.
- Sensor calibration, installation depth/orientation and site-specific thresholds will affect field performance.
- The supplied hardware images contain two component variants; a single final BOM should be frozen before fabrication.

- The example dashboard contains illustrative operational values and prototype model metrics; these should not be treated as measured field results unless independently validated.
- Alert thresholds should be configured with domain experts and local authorities and tested against historical events.

## **27. Future Enhancement Roadmap**

- Expand the labeled landslide-event dataset and perform site-specific model validation.
- Compare XGBoost with additional tree-based and temporal models under the same evaluation protocol.
- Introduce probability calibration and uncertainty reporting.
- Add automated satellite ingestion and reproducible SAR/NDVI preprocessing.
- Implement adaptive sampling so sensor nodes can increase measurement frequency during elevated risk.
- Add multilingual and low-literacy warning interfaces for field deployment.
- Introduce SMS/voice or other configured notification channels where required.
- Develop digital-twin or terrain-simulation views for scenario analysis.
- Improve enclosure, sensor mounting and long-term environmental protection through field trials.
- Implement secure firmware update and device lifecycle management.

## **28. Conclusion**

The supplied design describes a comprehensive multi-source landslide early-warning architecture that connects field sensing, LoRa communication, backend services, AI/ML prediction, GIS visualization and operational alerting. The system is centered on an ESP32-S3/LoRa field node and a 12-feature analytical framework covering rainfall, soil moisture, ground tilt, vibration, air quality, temperature, SAR change, NDVI, slope, elevation, land cover and historical landslides. The XGBoost engine produces a risk estimate that is represented as a 0–100 score and four operational classes, while explainability and GIS layers provide decision context.

The next critical engineering step is to move from a visually complete prototype architecture to a validated, reproducible implementation: freeze the final hardware BOM, define sensor calibration procedures, build the labeled training dataset, document the model evaluation protocol, finalize the API schemas, implement secure backend services, and conduct controlled and field tests. Once these elements are documented, the platform can serve as a strong technical foundation for a deployable landslide-monitoring and early-warning system.

## 29. References / Source Materials

- User-supplied system architecture flowchart: Satellite / IoT + LoRa / Weather → Data Fusion → AI/ML Engine → Risk Score → GIS Risk Visualization → Early Warning → Authority / Field Team / Web Dashboard.
- User-supplied hardware specification table and PCB/system component illustration.
- User-supplied circuit architecture showing solar charging, battery, protection, DC-DC conversion, ESP32-S3, sensors, LoRa and status indicators.
- User-supplied dashboard mock-ups covering regional risk, GIS visualization, AI prediction, explainability, LoRa connectivity, weather context and early-warning functions.
- User-specified backend architecture and REST API list: /api/sensor-data, /api/predict, /api/dashboard/{zone\_id}, /api/alerts, /api/alerts/{zone\_id}, /api/zones and /api/login.

**End of Technical Research Report**